Building Violations Detection & Reconciliation System (BVDRS)

Business Requirements Document (BRD) — Phase 1: Capital City & Village Peripheries

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Project:	Building Violations Detection and Reconciliation System (BVDRS)
Client:	Governorate of Al-Wadi, Arab Republic of Egypt
Sponsor:	Eng. Hesham, Secretary General of the Governorate
Evaluating Firm:	StrategizeIT Management & Technology Consulting Practice
Role:	Technical Business Analyst Candidate
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Prepared strictly in accordance with official case study facts, statutory parameters, and systems engineering standards.

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1. Document Control and Governance Sign-Off

Document ID	BVDRS-BRD-ALW-2026-V1.0	**Target Release**	Phase 1: Capital City & Adjacent Villages
Version / Status	v1.0 — Draft for Technical Review & Defense	**Primary Standard**	BABOK v3 / ISO-IEC-IEEE 29148 / BPMN 2.0
Author / Role	Muhammad Amjad Karara (Technical BA)	**Spatial Standard**	EPSG:32636 (WGS 84 / UTM Zone 36N)
Executive Sponsor	Eng. Hesham (Secretary General)	**Hosting Target**	On-Premises Governorate Data Center (Oracle 11g)

Table 1: Document Metadata and Baseline Control

Ver.	Date	Author	Summary of Baseline Scope and Revisions
v1.0	22-Oct-2026	Muhammad Amjad Karara	Initial Assessment Deliverable: Calibrated draft baseline (49 FRs, 16 NFRs, 15 BRs, 12 Data Entities, 10 Open Issues) prepared for technical defense.

Table 2: Version Revision History

Stakeholder	Official Title / Department	Governance Sign-Off Scope	Approval Status
Eng. Hesham	Secretary General (Sponsor)	Overall mandate, committee scaling [OI-01], and budget [OI-03].	Pending Sign-off
Eng. Mona	Head of Engineering & Planning	Technical validation, spatial zones, and Fast-Track policy [OI-02].	Pending Sign-off
Mrs. Nagwa	Director of Financial Affairs	Statutory tariffs, street overrides, 25% deposit, and arrears [OI-06].	Pending Sign-off
Mr. Adel	General Legal Counsel	30-day appeal moratorium lock, legal notices, and rule parametrization.	Pending Sign-off
Mr. Walid	IT Infrastructure Manager	On-premises hosting, Oracle 11g migration, security, and offline sync.	Pending Sign-off
Eng. Ahmed	Central GIS Unit Lead	Quarterly raster pipeline, WGS 84 transformation [OI-04], and UPIDs.	Pending Sign-off
Mr. Sayed	Technical Centers GM	Assisted-digital clerk workflows, POA validation, and archive scope [OI-10].	Pending Sign-off

Table 3: Mandatory Governance Sign-Off Matrix

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2. Executive Summary

2.1. Context and Municipal Enforcement Problem

Al-Wadi Governorate (population 3.2M; 8 cities, 200+ villages) faces severe illegal construction encroachment on agricultural land alongside an acute operational bottleneck in processing statutory reconciliation applications. Phase 1 of the **Building Violations Detection and Reconciliation System (BVDRS)** targets the capital city, **Madinat Al-Wadi**, and its adjacent rural envelope across 6 district offices and 4 technical centers.

The baseline operating environment suffers from critical structural failures:

- **Detection Latency & Report Loss:** Quarterly 50 cm satellite imagery requires ∼6 weeks of manual comparison by a 4-person GIS team. Paper dispatch to 25 field engineers results in unquantified report attrition (*"some reports never come back"*), leaving the central GIS map permanently outdated.
- **Fragmented Reconciliation Intake:** An unresolved backlog of ∼38,000 files grows by ∼1,500 monthly applications. Records are scattered across paper dossiers and 10 disconnected Excel spreadsheets (4 technical centers, 6 districts), while 3 municipal committees review only ∼80 files per weekly sitting (240 files/week total).

2.2. Quantitative Capacity Deficit Analysis

The Executive Sponsor (Eng. Hesham) stated a target objective to resolve the entire backlog within 6 months (26 weeks) of go-live [OI-01]. Clearing 38,000 backlog files plus 9,000 incoming applications over 26 weeks requires adjudicating **1,808 files/week**. Against current capacity (240 files/week), the Governorate operates at an **86.72% throughput deficit** ($7.53\times$ gap), causing the backlog to expand by **+460 files/month**. Furthermore, requiring 100% physical site visits across 47,000 files imposes an infeasible load of ∼**15 site inspections/engineer/day** across the 25 field engineers.

2.3. Target BVDRS Architecture and Governance Alignment

BVDRS establishes a centralized, spatial-first digital platform built on five pillars:

1. **Automated Change Detection:** Ingests quarterly 50 cm rasters to isolate candidate polygons within $\leq$48 h [TBC], filtering out Oracle 11g licensed buildings and active reconciliation parcels.

2. **Hybrid Assisted-Digital Intake:** Unifies citizen web self-service and technical center clerk entry (supporting Power of Attorney filings) on one database, generating Unique Property Identifiers (UPID) for uncadastered parcels.
3. **Spatial Pre-Screening & Triage:** Executes topological checks ($\leq$10 s) against urban boundaries, Nile buffers, antiquities, and airport height surfaces, triaging zero-conflict dossiers for expedited committee desk review [OI-02] without automated computer approval.
4. **Offline-First Mobile Inspection:** Equips 25 engineers with tablets featuring cached vector tiles, SQLite storage, and in-app GPS-locked photo capture with automated background synchronization.
5. **Dynamic Valuation, Moratorium Lock & Utility API:** Automates zone/street pricing, tracks 36-month instalments, enforces a 30-day appeal `STAY_OF_EXECUTION` lock, and exposes a read-only REST API for utility verification.

Consulting Verdict: Software alone cannot overcome an 86.7% human capacity deficit. Achieving the 6-month target requires pairing BVDRS triage with executive authorization of $\sim$20 additional weekly committee sittings and ratifying the Fast-Track desk review policy [OI-01, OI-02].

3. Business Objectives and Key Performance Indicators (KPIs)

3.1. Operational Capacity Modeling and Diagnostic Prose

To evaluate the Executive Sponsor's target of clearing the backlog within 6 months (26 weeks) [OI-01], a formal capacity model was established from case brief parameters (Table 4). Adding 6 months of incoming intake ($6 \times 1,500 = 9,000$) to the 38,000 pending backlog yields a total 26-week workload of **47,000 dossiers**, requiring a clearance velocity of **1,808 files/week** ($\sim$7,833/month). Conversely, the 3 active committees sitting once weekly ($\sim$80 files/sitting) adjudicate only **240 files/week** (1,040/month), clearing just **6,240 files (13.28%)** in 26 weeks. This reveals an **86.72% operational throughput deficit** ($7.53\times$ capacity multiplier), with the net backlog actively growing by **+460 files/month** (+2,760 files over 6 months). Simultaneously, mandating physical inspections for all 47,000 cases across 25 engineers requires 72.3 visits/engineer/week, or $\sim$**15 physical inspections/engineer/day** (assuming a 5-day work week)—an operational impossibility in unmapped rural zones. Closing this gap requires pairing BVDRS spatial triage [OI-02] with administrative authorization of $\sim$20 additional weekly committee sessions ($1,808 \div 80 \approx 23$ sessions total) [OI-01].

Baseline Parameter	Value	Source	Derived Capacity Metric	Value	Implication
Pending Backlog / Monthly Intake	38k / 1.5k	Brief §2.3	26-Week Total Workload	47,000 files	38k + 9k inflow
Target Clearance Window	26 weeks	Hesham [OI-01]	Required Weekly Velocity	1,808 files/wk	$\sim$23 sittings/wk
Active Committees / Cadence	3 / 1 wk	Brief §2.3	Status Quo Weekly Capacity	240 files/wk	1,040 files/mo
Capacity per Committee Sitting	$\sim$80 files	Brief §2.3	Operational Throughput Deficit	86.72% ($7.53\times$)	+460/mo growth
Field Engineers (6 Districts)	25 staff	Brief §2.3	Daily Engineer Visit Load	$\sim$15 visits/day	Physically infeasible

Table 4: Baseline Operational Parameters and Derived Capacity Deficit Metrics

3.2. Business Objectives and Measurable KPIs

ID	Business Objective	Statement of Strategic Business Purpose	Strategic Driver
BO-01	Backlog & Intake Control	Provide triage workflows to process 1,500 monthly files and clear the 38k backlog [OI-01].	Operational Recovery
BO-02	Rapid Change Detection	Compress quarterly satellite change extraction from 6 weeks to $\leq$48 h [TBC].	Urban Containment
BO-03	Digital Lifecycle Integrity	Unify intake, inspection, voting, and certification into one auditable system, ending paper loss.	Governance & Trust
BO-04	Municipal Revenue Assurance	Automate zone/street valuation, capture 25% deposits, and monitor 36-month instalments.	Fiscal Sustainability
BO-05	Geospatial Single Truth	Standardize layers on WGS 84 / UTM 36N and generate UPIDs for un-cadastered properties.	Spatial Accuracy
BO-06	Utility API Verification	Provide instant online QR/REST certificate verification for utilities, replacing postal mail.	Inter-Agency Control

Table 5: Strategic Business Objectives Register

ID	KPI Name	Baseline (AS-IS)	Target (TO-BE)	Measurement Methodology	BO
KPI-01	Change Pipeline Time	∼6 weeks manual	≤ 48 h from raster [TBC]	Raster ETL job execution timestamps.	BO-02
KPI-02	Field Report Return	Unquantified loss	100% digital sync	Tablet sync count vs. dispatched tickets (30-day baseline).	BO-03
KPI-03	Committee Velocity	240/wk (1,040/mo)	Stg 1: 1.5k/mo; Stg 2: 1,808/wk	Weekly electronic docket decision audit logs [OI-01].	BO-01
KPI-04	Spatial Screening Time	Unmeasured manual	≤ 10 s per dossier [TBC]	Automated spatial intersection query timer.	BO-05
KPI-05	Inspection Sync Latency	Untracked courier	≤ 24 h from site visit [TBC]	Timestamp delta between field capture and DB sync.	BO-03
KPI-06	Instalment Monitoring	0% digital tracking	100% tracked & alerted	Ratio of overdue accounts flagged with SMS alerts.	BO-04
KPI-07	Utility API Latency	Multi-week postal mail	≤ 3 s via REST API [TBC]	API gateway response time logs.	BO-06
KPI-08	Geodetic Alignment	Shifted tens of metres	Error ≤ 0.3 m [TBC / OI-04]	RMSE against Survey Authority ground control points.	BO-05

Table 6: Measurable Key Performance Indicators (Baseline vs. Target)

4. Project Scope

4.1. Geographical Phasing and In-Scope Functional Modules

- **Phase 1 (Immediate Scope):** Capital city (**Madinat Al-Wadi**) and adjacent rural villages across 6 district engineering offices and 4 municipal technical centers.
- **Future Rollout (Phase 2/3):** Remaining 7 cities and 200+ villages across Al-Wadi Governorate, plus live National Portal SSO and Civil Registry API handshakes.

In-Scope Phase 1 Architecture (9 Core Modules): (1) MOD-01: Quarterly 50 cm satellite change detection ($\leq$48 h [TBC]) filtered against Oracle 11g permits; **(2) MOD-02:** Offline-first tablet inspection app for 25 engineers with SQLite and GPS-locked photos; **(3) MOD-03:** Hybrid intake (citizen web portal + assisted-digital clerk UI) with Power of Attorney (POA) recording and UPID generation; **(4) MOD-04:** Automated spatial screening ($\leq$10 s) and Fast-Track desk triage [OI-02]; **(5) MOD-05:** Multi-committee digital dockets, 15-day deficiency timer [TBC], and 30-day `STAY_OF_EXECUTION` appeal lock; **(6) MOD-06:** Dynamic fee valuation (Zones A–D, street overrides), 25% deposit, 36-month instalments, and configurable parameter admin; **(7) MOD-07:** QR-encrypted certificates and read-only Utility Verification REST API; **(8) MOD-08:** Role-based executive GIS and financial dashboards; **(9) MOD-09:** Legacy ETL migration (Oracle 11g, 10 Excel sheets) and WGS 84 reprojection.

4.2. Explicit Out-of-Scope Boundaries and Analytical Justifications

ID	Excluded Capability	Requestor	Architectural & Governance Justification
OOS-01	Utility Meter Installation Field Crew Scheduling	Eng. Khaled (Electricity Co.)	**Separation of Concerns:** BVDRS is a municipal regulatory platform, not an external Workforce Management (WFM) system for commercial utilities. Scheduling utility crews breaches organizational boundaries and exposes the Governorate to liability for connection delays. Resolved via read-only `REST Verification API`.
OOS-02	Weekly Satellite Imagery Procurement	Eng. Hesham (Sponsor)	**Contractual Constraint:** Pipeline supports any raster cadence, but procuring weekly satellite passes is an external budget decision [OI-03]. Phase 1 operates strictly on the existing quarterly 50 cm imagery contract.
OOS-03	Full Digitization of 40k Closed Paper Archive	Mr. Sayed (Tech Centers)	**Resource Prioritization:** Phase 1 focuses strictly on the ∼38,000 active pending backlog files [OI-10]. Bulk scanning of closed historical paper dossiers from the prior law is deferred; files are digitized on-demand only upon active legal appeal.
OOS-04	Live API Integration with Civil Registry	Mr. Walid (IT Director)	**Ministerial Lead Time:** Inter-ministerial approval requires multi-month bureaucratic clearance [DEP-02 / OI-09]. Phase 1 uses clerk physical National ID inspection and 14-digit regex validation.
OOS-05	Physical Demolition & Security Logistics	District Enforcement	**Administrative vs. Physical Mandate:** BVDRS governs administrative legal notices and 30-day appeal moratorium locks. Physical machinery dispatch and police coordination remain external field operations.

Table 7: Explicit Out-of-Scope Exclusions and Defensible Justifications

5. Stakeholder Analysis

ID	Stakeholder	Role / Dept	Core Mandate & Stated Interests	Key Operational Tension / Risk	Grid Quadrant
SH-01	Eng. Hesham	Secretary General (Sponsor)	Resolve 38k backlog in 6 months [OI-01]; eliminate center queues via portal; executive map dashboard.	86.7% committee deficit; tension with walk-in demographics and quarterly imagery.	High Power / High Interest
SH-02	Eng. Mona	Head of Planning	Mandates 100% physical inspection [OI-02]; automated exclusion buffers; citizen map-pinning.	100% site visits impose ∼15 visits/eng/day across 25 engineers.	High Power / High Interest
SH-03	Eng. Ahmed	GIS Unit Lead	Manages 4 GIS staff; flags quarterly imagery, Red Belt datum shift, and 60% un-cadastered rural parcels.	6-week manual diff latency; false boundary rejections from datum shifts [OI-04].	Low Power / High Interest
SH-04	Mr. Sayed	Tech Centers GM	Manages 4 centers; champions elderly/illiterate and lawyer/POA assisted intake; inquiry tracking.	Fragmented paper and 10 Excel sheets; resists online-only exclusion of walk-ins.	Low Power / High Interest
SH-05	Mr. Walid	IT Director	Enforces on-premises hosting (Oracle 11g, ArcGIS); flags village internet outages and Civil Registry delay.	Strict data residency; rural offline needs; ministerial approval lead times [OI-09].	High Power / High Interest
SH-06	Mrs. Nagwa	Finance Director	Enforces Zones A–D tariffs, street overrides, 25% non-refundable deposit, and 36-month instalments.	Unmonitored instalment arrears; refund disputes on rejected files [OI-06].	High Power / High Interest
SH-07	Mr. Adel	Legal Counsel	Enforces 30-day appeal moratorium lock, formal demolition notice, and configurable rules [A-06].	Premature demolition risks litigation; frequent statutory amendments.	High Power / High Interest
SH-08	Eng. Khaled	Electricity Co.	Demands online certificate check before meter install; requested utility crew dispatch scheduling.	Scope creep into commercial utility WFM logistics (excluded via OOS-01).	Low Power / Low Interest
SH-09	Field Staff	25 District Engs	Need offline tablet navigation, digital forms, and end to paper map double-entry.	Half of field time lost locating rural buildings; cellular dead zones.	Low Power / High Interest
SH-10	Committees	3 Review Panels	Need organized digital dockets, spatial summaries, and rapid quorum voting.	Capped at ∼80 files/sitting (240/wk) vs. 1,808/wk required velocity.	Low Power / High Interest
SH-11	The Governor	Provincial Exec	Mandates provincial backlog resolution; reviews high-level spatial and fiscal KPIs.	Political exposure if 6-month target slips without resource scaling.	High Power / Low Interest

Table 8: Stakeholder Register and Power/Interest Classification

5.1. RACI Governance Matrix

Functional Lifecycle Area	Hsh	Mna	Ahmd	Syd	Wld	Ngw	Adl	Cmte	Engs
Satellite Imagery Ingestion & Detection	I	C	**R/A**	I	C	I	I	I	C
Field Violation Inspection & Mobile Sync	I	**A**	C	I	I	I	I	I	**R**
Citizen & POA Application Intake	I	C	I	**R/A**	C	C	I	I	I
Spatial Screening & Fast-Track Triage	I	**A**	**R**	I	C	I	C	C	I
Committee Docket Review & Adjudication	I	C	I	I	I	I	C	**R/A**	C
Appeals Lifecycle & Moratorium Lock	I	C	I	I	I	I	**A**	**R**	I
Fee Valuation, Down Payment & Instalments	I	I	I	C	I	**R/A**	C	I	I
Certificate Issuance & Utility REST API	I	**A**	I	**R**	C	C	C	I	I
Legacy Data Migration (Oracle/Excel/Datum)	I	C	**R**	C	**A**	C	I	I	I
Executive Dashboards & BI Reporting	**A**	C	C	C	**R**	C	I	I	I

Table 9: BVDRS Operational RACI Matrix (R: Responsible, A: Accountable, C: Consulted, I: Informed)

6. AS-IS Process Analysis

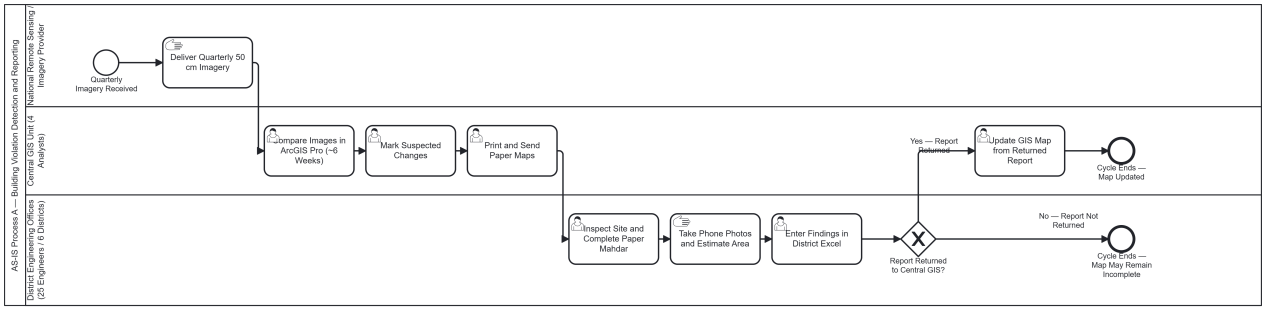
6.1. Baseline Workflows: Process A (Detection) and Process B (Reconciliation)

- **Process A (AS-IS Detection):** (1) Quarterly 50 cm WGS 84 satellite imagery arrives; (2) 4 GIS analysts spend ∼6 weeks manually comparing rasters in ArcGIS Pro; (3) Suspected points are printed on paper maps and couriered to 6 districts; (4) 25 field engineers visit sites, handwrite reports (*Mahdar*), take unverified personal phone photos, and estimate areas; (5) Findings are typed into 6 local district Excel sheets, while an unquantified share of reports never returns to GIS (*"some reports never come back"*).
- **Process B (AS-IS Reconciliation):** (1) Citizens or legal agents/lawyers submit paper dossiers at 4 technical centers; (2) Clerks verify paper checklists, log entries in 4 isolated center Excel files, and collect the 25% down payment; (3) Paper files move to districts where engineers spend half their field time locating rural buildings (60% lack cadastre) and manually cross-check paper restriction maps; (4) 3 committees meet weekly (∼80 files/sitting = 240/wk) to Approve, Reject, or Request Docs; (5) Finance manually calculates fees (Area × Zone), deducts 25%, and issues paper vouchers for lump-sum or 36-month instalments (with zero arrears tracking); (6) Paper certificates are mailed to utilities; rejected applicants file paper appeals within 30 days.

ID	Pain Point	Step	Root Cause and Operational Impact	Severity
P-01	6-Week Detection Latency	Proc A.2	Manual visual raster comparison allows illegal structures to finish before dispatch.	Critical
P-02	Unquantified Report Loss	Proc A.5	Paper routing results in reports never returning to GIS, blinding leadership.	High
P-03	Unverified Phone Photos	Proc A.4	Personal smartphone photos lack GPS locks or hashes, inviting legal challenges.	Medium
P-04	10 Siloed Excel Sheets	Proc B.2	4 center + 6 district spreadsheets create duplicates and zero central visibility.	Critical
P-05	Geodetic Datum Shift	Proc B.3	Red Belt vs. WGS 84 mismatch shifts layers by tens of metres, causing false flags.	Critical
P-06	86.7% Committee Deficit	Proc B.4	240 files/wk capacity vs. 1,808/wk required causes +460 files/mo backlog growth.	Critical
P-07	Untracked Arrears	Proc B.5	Zero digital tracking for 36-month instalments risks severe municipal revenue loss.	High
P-08	Postal Certificate Fraud	Proc B.6	Mailed utility letters suffer delays and forgery risks without online verification.	High

Table 10: AS-IS Critical Pain Points Register

Process A: AS-IS Detection Workflow (BPMN 2.0)



Process B: AS-IS Reconciliation Workflow (BPMN 2.0)

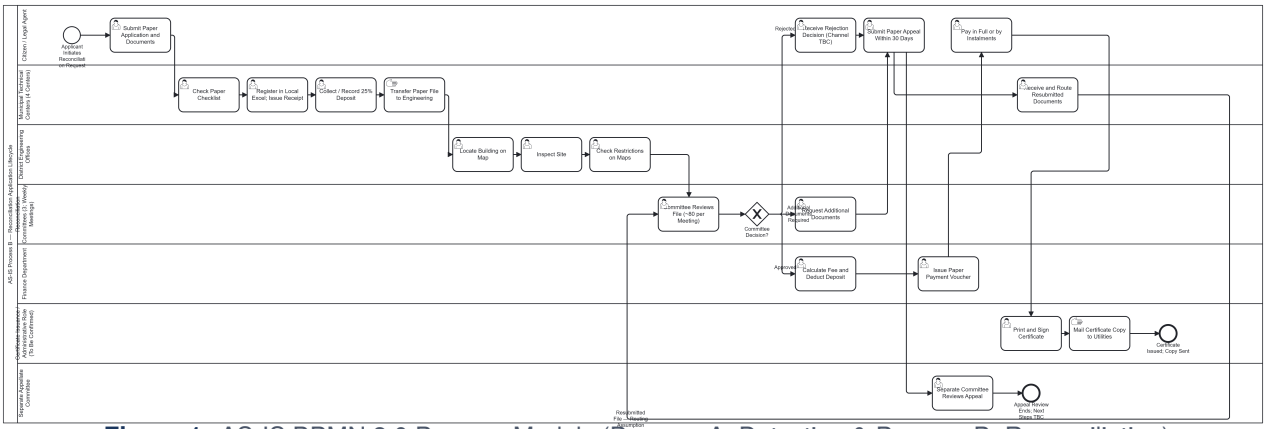


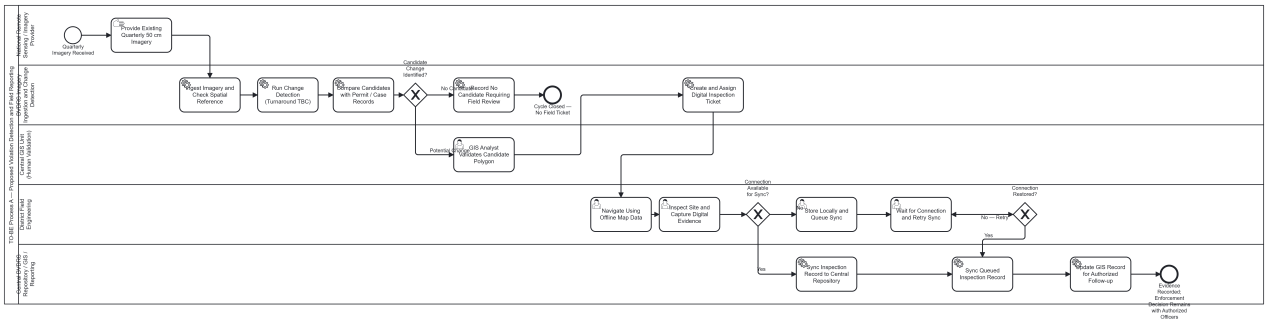
Figure 1: AS-IS BPMN 2.0 Process Models (Process A: Detection & Process B: Reconciliation)

7. TO-BE Process Design

7.1. Target Workflows: Process A (Rapid Detection) and Process B (Integrated Reconciliation)

- **Process A (TO-BE Detection):** (1) Quarterly 50 cm rasters ingest into the CV change pipeline, extracting candidate polygons in $\leq$48 h [TBC]; (2) System auto-filters polygons against migrated Oracle 11g permits and active reconciliation files; (3) GIS analysts validate clusters and dispatch digital tickets to district queues; (4) Field engineers inspect sites offline using tablets (cached vector basemaps, SQLite, GPS-locked in-app camera, digital forms); (5) Tablets auto-sync upon connectivity, updating executive heatmaps and generating violation notices.
- **Process B (TO-BE Reconciliation):** (1) Citizens or lawyers/agents (with validated POA) apply via web portal or assisted-digital clerk counters, pinning the building to generate a UPID; (2) Instant spatial pre-screening checks prohibition buffers ($\leq$ 10 s) *prior* to payment invoicing to prevent refund disputes [OI-06]; (3) Applicant uploads mandatory docs and pays the 25% down payment via e-payment gateway; (4) **Triage Gateway:** Compliant zero-conflict files route to Fast-Track committee desk review [OI-02] (toggle OFF by default), while complex/unverified cases dispatch to mobile field inspection; (5) Committees vote electronically (*Approve*, *Reject*, or *Request Docs* with 15-day timer [TBC]); (6) Valuation engine computes net fees (Area $\times$ Zone/Street Rate $-$ 25%) and schedules up to 36-month instalments; (7) Encrypted QR certificate issues [OI-08] and is verified by utilities via read-only REST API.
- **Exception Paths: (A) Pre-Payment Prohibition Block:** Halts fee collection for parcels inside Nile/antiquities zones; **(B) 30-Day Appeal Moratorium:** Rejections trigger an immutable `STAY_OF_EXECUTION` lock barring demolition orders during the 30-day appeal window and appellate review; **(C) Arrears Suspension:** Overdue instalments trigger SMS reminders, penalty flags ($T + 30$), and transition certificate status to `SUSPENDED` ($T + 60$ [TBC / OI-06]) on the utility API.

Process A: TO-BE Automated Detection and Mobile Inspection (BPMN 2.0)



Process B: TO-BE Reconciliation Lifecycle and Moratorium Lock (BPMN 2.0)

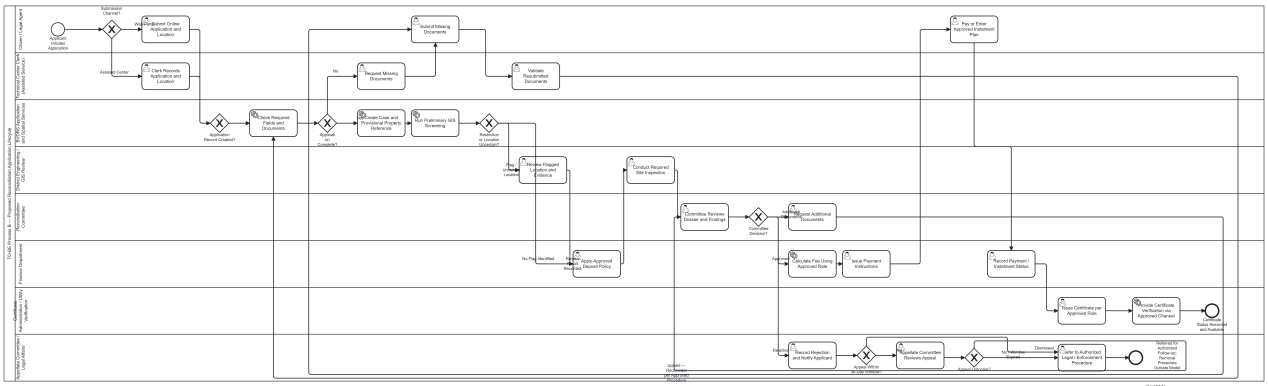


Figure 2: TO-BE Target State BPMN 2.0 Process Models (Process A: Detection & Process B: Reconciliation)

8. Gap Analysis (AS-IS vs. TO-BE)

8.1. Multi-Dimensional Gap Matrix and Closure Governance

Table 11 maps baseline deficiencies across Process, System, Data/GIS, Policy, and People dimensions to their target BVDRS resolution mechanisms. While ten operational gaps are closed directly by software automation, closing **GAP-02** and **GAP-11** requires executive governance ratification of the Fast-Track desk review toggle and committee session scaling ($\sim$20 additional weekly sittings) [OI-01, OI-02].

ID	Dimension	AS-IS Baseline State	TO-BE Target State	Resolution Mechanism (How Closed)
GAP-01	Process	6-week manual visual raster comparison in ArcGIS Pro.	Automated change extraction within ≤ 48 h [TBC].	CV raster diff pipeline filtering changes against Oracle 11g building permits (`FR-DET-002/003`).
GAP-02	Process	100% physical inspection across all 47,000 files.	Automated spatial triage of low-risk compliant files.	Fast-Track Triage routes zero-conflict files to committee desk review; toggle OFF by default [OI-02].
GAP-03	System	10 isolated Excel files across 4 centers and 6 districts.	Unified enterprise database (Oracle 11g / Spatial).	Centralized transactional and spatial repository serving all portal, clerk, mobile, and committee UIs.
GAP-04	System	Postal letters mailed to utilities; forgery risks.	Real-time digital verification gateway (≤ 3 s).	Read-only RESTful Verification API and cryptographic QR code scanner for utilities (`FR-UTL-001/003`).
GAP-05	Data/GIS	Legacy Red Belt layers shifted tens of metres.	Single geodetic standard: WGS 84 / UTM Zone 36N.	Spatial ETL reprojection using Survey Authority parameters (≤ 0.3 m target error [TBC / OI-04]).
GAP-06	Data/GIS	60% of informal/rural buildings lack cadastral parcels.	100% spatial indexing and unique property tagging.	Algorithmic Unique Property Identifier (UPID) and visual placeholder generation at intake (`FR-INT-005`).
GAP-07	Data/GIS	Unquantified loss of paper field inspection reports.	100% tamper-proof digital field report sync.	Offline-first tablet app with in-app GPS-locked photos, SHA-256 hash, and auto-sync (`FR-FLD-001/006`).
GAP-08	Policy	Refund disputes over non-refundable 25% deposit.	Upfront spatial pre-screening before fee invoicing.	Instant prohibition check precedes payment voucher generation + signed digital disclaimer [OI-06].
GAP-09	Policy	Undefined rates in new urban extension areas.	Parameterized valuation preventing fee guessing.	System halts billing and flags `AWAITING_PRICING` until official pricing decree entry [OI-05].
GAP-10	Policy	Premature demolition risk during 30-day appeal window.	Automated legal moratorium state-machine lock.	Database lock (`STAY_OF_EXECUTION`) blocks demolition orders during 30-day window and active appeals.
GAP-11	People	240 files/wk capacity vs. 1,808 required (86.7% gap).	Clearance velocity aligned with 6-month target.	Fast-Track desk triage paired with executive authorization of $\sim$20 extra committee sittings/wk [OI-01].
GAP-12	People	Half of field time lost locating rural buildings.	Streamlined mobile navigation and digital reporting.	Offline vector basemaps, GPS pin routing, and single-entry mobile forms (`FR-FLD-001/005`).

Table 11: Multi-Dimensional Gap Analysis Matrix

9. Functional Requirements

ID	Requirement Statement (The system SHALL...)	Pri	Source	Testable Acceptance Criteria
MOD-01: Automated Satellite Change Detection & Ingestion (7 FRs)				
FR-DET-001	Ingest quarterly 50 cm GeoTIFF rasters in EPSG:32636 (WGS 84 / UTM 36N).	Must	Eng. Ahmed	Ingests without warping; logs date and resolution.
FR-DET-002	Run automated raster diff against prior quarter within ≤ 48 h [TBC].	Must	Hesham [OI-03]	Extracts candidate change vectors in ≤ 48 h.
FR-DET-003	Filter candidate polygons against migrated Oracle 11g building permits.	Must	Brief; Walid	Permitted footprints auto-excluded from violation queue.
FR-DET-004	Filter candidate polygons against active/approved reconciliation dockets.	Must	Eng. Mona	Active reconciliation parcels excluded from ticketing.
FR-DET-005	Compute footprint area (m^2) and perimeter vertices for unpermitted polygons.	Should	Eng. Ahmed	Vector attributes store area (m^2) and coordinates.
FR-DET-006	Provide ArcGIS Pro-compatible QA workbench to verify candidate clusters.	Must	Eng. Ahmed	Analysts approve, split, or dismiss clusters pre-dispatch.
FR-DET-007	Support ingestion of localized drone orthomosaics for Phase 2 hotspots.	Could	Eng. Ahmed	Renders localized high-res orthomosaics over base map.
MOD-02: Offline-First Mobile Field Inspection Application (7 FRs)				
FR-FLD-001	Operate offline on tablets using local SQLite DB and cached vector tiles.	Must	Walid; Engs	Full map navigation and form entry with zero signal.
FR-FLD-002	Route verified inspection tickets to engineers by district boundary.	Must	Eng. Mona	Tickets populate assigned district engineer queue.
FR-FLD-003	Capture in-app photos with GPS lock, timestamp, and SHA-256 hash.	Must	Eng. Mona	Blocks gallery uploads; embeds verified GPS/time hash.
FR-FLD-004	Allow on-site GPS vertex capture to refine building polygon geometry.	Must	Mona; Ahmed	Field GPS vertices update footprint within device tolerance.
FR-FLD-005	Provide digital violation form (*Mahdar*: structure, floors, usage).	Must	Brief §2.2.A	Blocks submission until mandatory fields are valid.
FR-FLD-006	Capture on-site stylus signature of inspector and property representative.	Should	Legal (Adel)	Generated PDF embeds timestamped digital signatures.
FR-FLD-007	Auto-synchronize reports and photos to central DB upon network reconnection.	Must	Mr. Walid	Syncs within 60 s of reconnect with 0% data loss.
MOD-03: Multi-Channel Intake & Assisted-Digital Portal (7 FRs)				
FR-INT-001	Provide standalone Governorate web portal for citizen filing and tracking.	Must	Hesham [OI-09]	24/7 citizen submission, upload, and status tracking.
FR-INT-002	Provide assisted-digital clerk interface sharing the unified core database.	Must	Mr. Sayed	Clerks log walk-in dossiers into identical core DB.
FR-INT-003	Support legal agent/lawyer filing with Power of Attorney (POA) metadata.	Must	Mr. Sayed	Validates POA number, notary office, and agent National ID.
FR-INT-004	Require interactive map-pinning to capture WGS 84 building coordinates.	Must	Eng. Mona	Map pin records lat/long and links district jurisdiction.
FR-INT-005	Generate Unique Property Identifier (UPID) and visual bounding placeholder.	Must	Eng. Ahmed	Assigns collision-free UPID to un-cadastered properties.
FR-INT-006	Validate mandatory document uploads and generate 25% deposit invoice.	Must	Nagwa; Mona	Blocks incomplete files; issues e-payment voucher.
FR-INT-007	Dispatch automated SMS/portal alerts with tracking ID and payment status.	Should	Mr. Sayed	Sends SMS within 30 s of submission and payment.
MOD-04: Automated Spatial Screening & Triage Engine (7 FRs)				
FR-SPC-001	Intersect pinned coordinates against official urban/village boundaries.	Must	Eng. Mona	Flags inside urban boundary vs. outside agricultural land.
FR-SPC-002	Intersect coordinates against Nile buffer, antiquities, and airport OLS.	Must	Eng. Mona	Returns boolean intersection and distance in ≤ 10 s [TBC].
FR-SPC-003	Route zero-conflict urban files to Fast-Track committee desk review [OI-02].	Should	Proposed; Mona	Policy toggle OFF by default; routes to expedited docket.
FR-SPC-004	Auto-generate mobile inspection tasks for complex or disputed cases.	Must	Eng. Mona	Non-fast-track files route to district tablet queues.
FR-SPC-005	Append automated 1-page Spatial Screening Summary to committee dockets.	Must	Derived (BA)	Displays coordinates, zone class, and buffer intersections.
FR-SPC-006	Transform legacy Egypt 1907 / Red Belt layers to WGS 84 / UTM 36N.	Must	Eng. Ahmed	Residual error ≤ 0.3 m [TBC: Subject to OI-04].
FR-SPC-007	Log all spatial screening queries, layer versions, and timestamps to audit DB.	Must	Walid; Adel	Immutable log records exact layer version evaluated.

Table 12: Functional Requirements Specification — Part 1 (Modules 1 to 4: FRs 1–28)

ID	Requirement Statement (The system SHALL...)	Pri	Source	Testable Acceptance Criteria
MOD-05: Committee Docket & Adjudication Management (6 FRs)				
FR-COM-001	Provide multi-committee electronic dockets supporting 3+ concurrent panels.	Must	Hesham; Panels	Concurrent independent docket access and voting.
FR-COM-002	Display unified single-screen dossier (owner/POA, safety report, GIS, photos).	Must	Derived (BA)	Consolidated view renders without external apps.
FR-COM-003	Record formal committee votes (Approve, Reject, Request Docs) + legal grounds.	Must	Brief §2.2.B	Enforces quorum sign-off and mandatory legal reason.
FR-COM-004	Trigger deficiency notice with configurable 15-day timer [TBC] on Doc Request.	Must	Brief §2.2.B	Sends SMS alert and holds file in deficiency queue.
FR-COM-005	Apply 30-day appeal timer and `STAY_OF_EXECUTION` lock upon rejection.	Must	Legal (Adel)	Programmatically blocks demolition orders during lock.
FR-COM-006	Provide Appellate Committee portal to adjudicate appeals and unlock enforcement.	Must	Legal (Adel)	Logs appellate ruling; unlocks demolition after expiry.
MOD-06: Dynamic Valuation, Invoicing & Settlement Engine (7 FRs)				
FR-VAL-001	Compute statutory fee by zone: A (1200), B (800), C (450), D (150) EGP/m^2.	Must	Mrs. Nagwa	Executes Area $\times$ Zone Rate with 100% precision.
FR-VAL-002	Apply specific street price overrides superseding general zone rates.	Must	Mrs. Nagwa	Street decree rate overrides zone tariff [OI-05].
FR-VAL-003	Halt billing and flag `AWAITING_PRICING` in unpriced extension areas.	Must	Mrs. Nagwa	Prevents fee guessing until official decree entry.
FR-VAL-004	Credit validated 25% down payment and compute net balance (lump-sum/discount).	Must	Mrs. Nagwa	Deducts 25% deposit; supports configurable cash discount.
FR-VAL-005	Generate amortization schedules up to 36 months (monthly/quarterly) [TBC].	Must	Mrs. Nagwa	Outputs due dates, principal, and payment references.
FR-VAL-006	Track receivables, send SMS alerts, and flag default at $T + 60$ days [TBC].	Must	Mrs. Nagwa	Default transitions certificate status to `SUSPENDED`.
FR-VAL-007	Provide Parameter Admin UI to update tariffs, timers, and rules without code.	Must	Legal (Adel)	Authorized admins modify rates/timers via UI [A-06].
MOD-07: Certificate & Utility Verification API (5 FRs)				
FR-UTL-001	Issue digitally signed certificates with encrypted 2D QR verification codes.	Must	Hesham; Khaled	PDF embeds tamper-proof cryptographically signed QR.
FR-UTL-002	Provide public web portal to verify certificates via QR scan.	Must	Sayed; Khaled	Displays UPID, address, status, and permitted floors.
FR-UTL-003	Expose secure, rate-limited read-only REST Verification API for utilities.	Must	Eng. Khaled	JSON response in ≤ 3 s [TBC] by UPID/National ID.
FR-UTL-004	Enforce data minimization on utility API (status and floors only; no PII/finance).	Must	Mr. Walid	Excludes financial and personal phone/bank records.
FR-UTL-005	Dynamically reflect `SUSPENDED` status on utility API upon instalment default.	Must	Nagwa; Khaled	Defaulted accounts immediately return `SUSPENDED`.
MOD-08: Executive Dashboards (3 FRs) & MOD-09: Legacy ETL Migration (4 FRs)				
FR-DSH-001	Provide Governor's Spatial Dashboard with district violation/reconciliation heatmaps.	Must	Eng. Hesham	Real-time GIS choropleth map and enforcement counters.
FR-DSH-002	Provide Committee Throughput Monitor tracking weekly clearance vs. 1,808 target.	Must	Hesham [OI-01]	Tracks sittings, approval rates, and backlog burndown.
FR-DSH-003	Provide Finance Receivables Dashboard tracking deposits and 30/60-day arrears.	Must	Mrs. Nagwa	Displays collected revenue and defaulted accounts.
FR-MIG-001	Migrate 2010–2019 building permits from legacy Oracle 11g database.	Must	Mr. Walid	Extracts permits with 0% loss for change filtering.
FR-MIG-002	Ingest and deduplicate active files from 10 legacy Excel workbooks (4+6 offices).	Must	Sayed; Walid	Deduplicates by National ID/address into unified DB.
FR-MIG-003	Reproject legacy spatial layers from Egypt 1907 / Red Belt to WGS 84 / UTM 36N.	Must	Eng. Ahmed	Applies certified shift parameters [OI-04] during ETL.
FR-MIG-004	Quarantine incomplete or duplicate legacy records into clerk exception queues.	Must	Sayed; Walid	Flags corrupt records for manual clerk remediation.

Table 13: Functional Requirements Specification — Part 2 (Modules 5 to 9: FRs 29–49)

10. Non-Functional Requirements

10.1. Quantitative Non-Functional Requirements Specification

Table 14 specifies the sixteen baseline Non-Functional Requirements (NFRs) governing performance, availability, sovereign security, and rural offline resilience.

ID	Requirement Title	Quantitative Target Specification	Verification Methodology
Category 1: Performance and Scalability			
NFR-PERF-001	Spatial Query Latency	Topological intersection queries complete in ≤ 2.0 s (simple) and ≤ 10.0 s (complex) [TBC].	Stress test: 50 concurrent spatial queries.
NFR-PERF-002	Repository Sizing	Supports $\geq 50,000$ active dossiers (38k backlog + intake), scalable to 100,000 without re-indexing.	Benchmark test with 50k seeded records.
NFR-PERF-003	User Concurrency	Sustains ≥ 150 concurrent staff and ≥ 500 portal citizens with page load ≤ 1.5 s.	Load test: 650 concurrent HTTPS sessions.
NFR-PERF-004	Utility API Latency	Verification gateway sustains ≥ 50 req/s with 95th-percentile response ≤ 500 ms (≤ 3 s max) [TBC].	API stress test at sustained 50 GET req/s.
Category 2: Availability, Disaster Recovery and Resilience			
NFR-AVAL-001	Production Uptime	Operational availability ≥ 99.5% during municipal working hours (08:00–18:00, Sat–Thu).	Monthly uptime monitoring logs.
NFR-AVAL-002	Recovery Point (RPO)	Maximum allowable data loss ≤ 1.0 hour upon catastrophic primary hardware failure.	Hourly Oracle archive redo log replication audit.
NFR-AVAL-003	Recovery Time (RTO)	Full production service restoration within ≤ 4.0 hours following data center failover.	Semiannual disaster recovery failover drill.
Category 3: Security, Privacy and Sovereign Data Residency			
NFR-SEC-001	Sovereign Hosting	All databases, spatial layers, and citizen PII reside strictly in the Governorate Data Center [A-01].	Network audit confirming zero external cloud egress.
NFR-SEC-002	Data Encryption	AES-256 encryption for data at rest (Oracle/SQLite); TLS 1.2/1.3 for all traffic in transit.	Cipher suite scan and storage encryption audit.
NFR-SEC-003	Access Control (RBAC)	Strict RBAC across Citizen, Clerk, Engineer, Committee, Finance, Legal, and Admin profiles.	Penetration test verifying privilege isolation.
NFR-SEC-004	Immutable Audit Log	Append-only logging for all votes, status changes, fee edits, and API queries (user, IP, timestamp).	DB trigger test blocking UPDATE/DELETE on audit log.
Category 4: Usability, Localization and Rural Connectivity Resilience			
NFR-USE-001	Arabic Localization	100% native Arabic Right-to-Left (RTL) interfaces across portal, clerk, committee, and mobile apps.	Clerk and engineer UAT localization sign-off.
NFR-USE-002	Accessibility Standard	Citizen web portal conforms to W3C WCAG 2.1 Level AA for elderly and low-vision accessibility.	Automated accessibility audit (≥ 90% score).
NFR-OFF-001	Offline Tablet Autonomy	Tablet app operates 100% offline, caching ≥ 500 tickets and district vector basemaps locally.	Airplane-mode field test of 10 full inspections.
NFR-OFF-002	Resilient Delta Sync	Background sync resumes automatically upon signal detection using delta compression.	Throttled link test (128 kbps, 50% packet loss).
NFR-NET-001	Low-Bandwidth UI	Technical center and portal forms submit reliably over degraded rural connections without data loss.	WAN emulation test confirming transactional integrity.

Table 14: Comprehensive Non-Functional Requirements Matrix

11. Business Rules

11.1. Statutory, Spatial, Financial, and Legal Business Rules

Table 15 codifies the fifteen mandatory business rules across four domains. Per Legal Counsel's mandate (`BR-LEGAL-004`), all tariff rates, grace timers, and thresholds are parameter-driven [A-06] to support legislative updates without code changes.

Rule ID	Rule Name	Formal Business Rule Logic	Source / Governance
Domain 1: Spatial Eligibility and Absolute Prohibitions (BR-SPAT)			
BR-SPAT-001	Urban Boundary Check	Violations are eligible ONLY IF inside the approved urban boundary (*El-Hayyez El-Omrani*) or village envelope; outside agricultural land is non-reconcilable.	Brief §2.1; Eng. Mona
BR-SPAT-002	Nile Buffer Prohibition	Any property intersecting the Nile River Protection Buffer (*Haram El-Nile*) is `ABSOLUTELY_PROHIBITED`; pre-screening halts fee collection [OI-06].	Brief §2.2.B; Eng. Mona
BR-SPAT-003	Antiquities Protection	Properties intersecting certified antiquities/heritage buffers are `NON_RECONCILABLE`; authoritative layer source confirmed via [OI-04].	Brief §2.2.B; Eng. Mona
BR-SPAT-004	Airport Height Limits	Building elevations exceeding civil aviation Obstacle Limitation Surfaces (OLS) are non-reconcilable for floors above the ceiling [OI-04].	Brief §2.2.B; Aviation Auth
Domain 2: Statutory Valuation and Fee Assessment (BR-VAL)			
BR-VAL-001	Zone Tariff Schedule	Base fee = Violating Area $\times$ Zone Rate: Zone A (City Center) = 1,200; Zone B (Planned) = 800; Zone C (Informal) = 450; Zone D (Villages) = 150 EGP/m^2.	Brief §2.4; Mrs. Nagwa
BR-VAL-002	Street Rate Override	Approved Pricing Committee street rates supersede general zone tariffs; system logs decree number and applies rate active at submission [OI-05].	Brief §2.4; Mrs. Nagwa
BR-VAL-003	Unpriced Extension Hold	Applications in unpriced urban extension areas are flagged `AWAITING_PRICING`; system strictly prohibits fee estimation or extrapolation.	Brief §2.4; Nagwa [OI-05]
BR-VAL-004	25% Down Payment	A mandatory 25% down payment is collected at intake (non-refundable upon rejection); spatial pre-screening runs prior to payment voucher issue.	Brief §2.4; Nagwa [OI-06]
BR-VAL-005	Cash Settlement Option	Supports full lump-sum settlement with a configurable cash discount parameter (disabled by default until executive decree ratification [A-06]).	Mrs. Nagwa; [A-06]
Domain 3: Instalment Amortization and Receivables (BR-INST)			
BR-INST-001	36-Month Maximum Term	Net balance (Total Fee $-$ 25%) may be amortized over up to 36 calendar months with monthly/quarterly frequency [TBC].	Brief §2.4; Mrs. Nagwa
BR-INST-002	Delinquency & Default	Accounts are flagged `IN_ARREARS` at $T + 30$ days past due and `IN_DEFAULT` at $T + 60$ days [TBC parameter per OI-06] with configurable penalties.	Mrs. Nagwa [OI-06]
BR-INST-003	Certificate Suspension	Transition to `IN_DEFAULT` automatically updates certificate state to `SUSPENDED` on the Utility Verification API, blocking meter installation.	Nagwa; Eng. Khaled
Domain 4: Appeals Moratorium and Legal Governance (BR-LEGAL)			
BR-LEGAL-001	30-Day Appeal Window	Rejected applicants have exactly 30 calendar days from confirmed notification delivery [OI-07] to submit a formal appeal (*Tazallom*).	Brief §2.5; Mr. Adel
BR-LEGAL-002	Stay of Execution Lock	Rejection activates an immutable `STAY_OF_EXECUTION` DB lock during the 30-day window and active appeal review, blocking demolition orders.	Mr. Adel (Legal Counsel)
BR-LEGAL-003	Demolition Notice	Upon appeal expiry or final dismissal, system unlocks enforcement, requiring legal sign-off and formal warning notice before police routing.	Brief §2.5; Mr. Adel
BR-LEGAL-004	Parameterized Rules	All tariffs, grace periods, appeal timers, and default thresholds are stored in editable parameter tables to support regulatory changes without recoding.	Mr. Adel [A-06]

Table 15: Comprehensive Business Rules Specification Matrix

12. Data and GIS Requirements

ID	Entity Name	Geometry	Key Attributes and Functional Purpose	Source / Module
ENT-01	`Building_Footprint`	Polygon	`UPID`, `Parcel_ID`, Area (m^2), Floors, Height, Usage, Status.	Imagery / Mobile GPS
ENT-02	`Cadastral_Parcel`	Polygon	`Parcel_ID`, Block, District, Registered Owner, Survey Ref.	Survey Auth (City)
ENT-03	`Restricted_Zone`	Polygon	`Zone_ID`, Type (Nile, Antiquities, Urban, OLS), Decree, Buffer.	Ministerial GIS Layers
ENT-04	`Violation_Cluster`	Point/Poly	`Cluster_ID`, Detection Date, Change Area, Baseline Raster Ref.	Change Pipeline (MOD-01)
ENT-05	`Dossier_Applicatio`	Relational	`Dossier_ID`, `UPID`, Owner National ID, Channel, Status.	Portal / Clerk (MOD-03)
ENT-06	`Legal_Agent_POA`	Relational	`Agent_ID`, `Dossier_ID`, Agent ID, POA Number, Notary Office.	Clerk Intake Validation
ENT-07	`Inspection_Report`	Point/Poly	`Report_ID`, `Dossier_ID`, Inspector ID, GPS Vertices, Photo Hashes.	Offline Tablet (MOD-02)
ENT-08	`Committee_Decision`	Relational	`Decision_ID`, `Dossier_ID`, Committee ID, Vote, Legal Justification.	Docket Module (MOD-05)
ENT-09	`Valuation_Record`	Relational	`Val_ID`, `Dossier_ID`, Zone Tariff, Street Override, 25% Deposit, Net Due.	Valuation Engine (MOD-06)
ENT-10	`Instalment_Plan`	Relational	`Plan_ID`, `Dossier_ID`, Term ($\leq$36m), Due Dates, Arrears Days, State.	Receivables Engine
ENT-11	`Appeal_Moratorium`	Relational	`Appeal_ID`, `Dossier_ID`, 30-Day Timer, `STAY_OF_EXECUTION` Lock.	Legal Module (Adel)
ENT-12	`Reconcil_Certificate`	Relational	`Cert_ID`, `Dossier_ID`, Encrypted QR Hash, Status (`ACTV/SUSP`).	Utility Gateway (MOD-07)

Table 16: Conceptual Data Entities and Spatial Feature Classes

12.1. Synthetic Parcel Indexing (UPID), Datum Transformation, and Legacy Migration

- **Cadastral Gap Resolution (UPID):** Because $\sim$60% of informal/rural properties lack formal cadastral parcels, map-pinning generates an algorithmic **Unique Property Identifier**: `UPID = [GOV]-[DIST]-[ZONE]-[CENTR`

(e.g., `ALW-01-C-E31045N30521-00842`). The synthetic bounding box acts strictly as a *visual UI placeholder*; definitive topological screening uses pinned coordinates refined by field GPS vertices [OI-05].

- **Geodetic Standardization:** Legacy maps in **Egypt 1907 / Red Belt** (EPSG:22992) shift by tens of metres against **WGS 84 / UTM Zone 36N** (EPSG:32636). All layers are reprojected to EPSG:32636 via Survey Authority transformation parameters with target residual error ≤ 0.3 m [TBC / OI-04].
- **Legacy Migration Plan:** (1) **Oracle 11g Permits (2010–2019):** Automated ETL extracts licensed footprints to filter change detection (`FR-DET-003`); (2) **10 Excel Workbooks (4 Centers + 6 Districts):** Deduplicated by National ID and address into the unified DB; (3) **Paper Archive Scope [OI-10]:** Phase 1 prioritizes the ∼38,000 active pending backlog dossiers; the ∼40,000 historical paper archive is digitized on-demand upon active legal appeal.

Geodetic Parameter	Legacy Baseline	Target Standard	Transformation Control
Coordinate System / Ellipsoid	Egypt 1907 Red Belt / Helmert 06	WGS 84 / UTM 36N (EPSG:32636)	Survey Authority 7-parameter / grid shift
Positional Alignment Accuracy	Shifted by tens of metres	Target residual error ≤ 0.3 m	Certified GCP validation [TBC / OI-04]

Table 17: Geodetic Datum Transformation Standards

13. Integration Requirements

13.1. Enterprise Integration Architecture and Utility Boundary

All system integrations are mediated through an on-premises API gateway within the Governorate Data Center to enforce security and sovereign data residency [A-01]. Per Scope Exclusion `OOS-01`, integration with external utilities (`INT-06`) is strictly restricted to read-only certificate verification returning only `UPID`, `Certificate_Status` (`ACTIVE`, `EXPIRED`, `SUSPENDED`), `Approved_Floors`, and `Issuance_Date` (excluding citizen financial records and PII), secured via mTLS and rate-limited to 100 req/min.

ID	Target System	Dir.	Protocol	Data Exchanged & Scope	Phase 1 Fallback Mechanism
INT-01	Governorate Citizen Web Portal	Bi-dir	HTTPS REST / JSON	Application intake, document uploads, map-pinning, and real-time status tracking [OI-09].	Standalone local auth + SMS OTP in Phase 1; National Portal SSO deferred to Phase 2.
INT-02	National Civil Registry (MOI)	Inbound	HTTPS REST / JSON	Automated National ID and legal name validation (Phase 2 target per ministerial approval).	**Phase 1 Baseline:** Clerk visual ID check + client-side 14-digit regex validation [DEP-02].
INT-03	Government E-Payment Gateway	Bi-dir	HTTPS REST / Webhook	25% down payment invoicing, instalment billing, and cryptographic payment receipts.	Asynchronous webhook retries; nightly batch settlement reconciliation file.
INT-04	ArcGIS Enterprise 11.1 Server	Bi-dir	GeoServices REST / JSON	Authoritative boundary/prohibition layers, candidate change vectors, and field polygons.	Local caching of boundary layers on app servers with daily sync replication.
INT-05	Legacy Oracle 11g Permits DB	Inbound	JDBC / Batch SQL	One-way extraction of 2010–2019 building permits for change detection filtering.	Isolated read-only staging replica protecting legacy database performance.
INT-06	Utilities Verification Gateway	Outbound	HTTPS REST / JSON	Read-only query by UPID/QR hash returning `ACTIVE/SUSPENDED` status and floors [OOS-01].	Distributed cache sustaining ≥ 50 req/s with mTLS and IP whitelisting.
INT-07	Government SMS Gateway	Outbound	HTTP POST / XML	Tracking IDs, 15-day deficiency alerts, 30-day appeal timers, and instalment reminders.	Persistent message queue with automatic retry; fallback to portal notifications.

Table 18: Enterprise System Integration Register

14. Reports and Dashboards

14.1. Decision-Oriented Reporting and Executive Spatial BI

Table 19 specifies the eight operational and executive reports. The Governor's Command Dashboard (`REP-01`) combines four core widgets: (1) **District Choropleth Map** (violation and reconciliation density), (2) **Through-**

put Funnel (*Intake* → *Screening* → *Inspection* → *Committee* → *Certified*), (3) **Live Revenue Ticker** (25% deposits, instalments collected, arrears), and (4) **Enforcement Matrix** (active 30-day `STAY_OF_EXECUTION` locks vs. unlocked demolition warrants).

ID	Report / Dashboard Title	Target Audience	Cadence	Decision Scope and Analytical Metrics
REP-01	Governor's Executive Spatial Dashboard	Governor; Secretary General	Real-time / Daily	Provincial GIS map showing violation heatmaps by district, reconciliation volume, revenue collected, and active appeal locks vs. demolition orders.
REP-02	Backlog Burndown & Capacity Monitor	Secretary General (Eng. Hesham)	Weekly	Tracks weekly adjudication velocity against the 6-month target (1,808/wk required vs. 240/wk baseline) to govern committee scaling [OI-01].
REP-03	Committee Productivity Docket	Committee Chairs; Secretary General	Weekly	Measures files decided per sitting, approval/rejection ratios, 15-day deficiency requests, and average cycle time per committee panel.
REP-04	Instalment Aging & Arrears Ledger	Finance Director (Mrs. Nagwa)	Daily / On-Demand	Segments 36-month receivables into aging buckets ($T+30$ arrears, $T+60$ default [TBC / OI-06]), flagging accounts with `SUSPENDED` certificates.
REP-05	Field Inspection Routing Docket	District Chief Engineers; Mona	Daily	Geographically clusters assigned mobile inspection tickets by village sector to optimize daily routes for the 25 field engineers.
REP-06	Spatial Restriction Incursion Heatmap	Central GIS Unit (Eng. Ahmed)	Quarterly / Monthly	Highlights illegal construction inside Nile buffers, agricultural land, and antiquities zones to direct targeted field enforcement.
REP-07	Statutory Financial Audit Journal	Finance Director; State Audit	Monthly	Reconciles 25% down payments, lump-sum cash settlements [A-06], instalment receipts, and e-payment gateway settlement batches.
REP-08	Citizen & POA Self-Service Tracker	Citizens; Legal Agents	Real-time	Public portal view displaying dossier stage, deficiency upload countdowns, payment receipts, and QR certificate status.

Table 19: Reports and Executive Dashboards Catalog

15. Assumptions, Risks, Constraints, and Dependencies

ID	Stakeholder Conflict	Parties	Defensible Resolution in BVDRS	Status
CON-01	6-Month Target vs. 240/wk Capacity	Hesham vs. Panels	Proved 86.7% deficit; deployed triage + recommended ∼20 extra sittings/wk.	Pending [OI-01]
CON-02	Weekly Detection vs. Quarterly Rasters	Hesham vs. Ahmed	Phase 1 baselined on quarterly 50 cm contract; drone ingestion in Phase 2.	Scoped [OI-03]
CON-03	Online-Only vs. Walk-In Citizens/POAs	Hesham vs. Sayed	Hybrid Assisted-Digital model: citizen web portal + clerk desk UI on one DB.	Resolved [A-02]
CON-04	100% Site Visits vs. Engineer Load	Mona vs. Capacity	Physical visit is baseline; Fast-Track desk triage built as policy toggle (OFF).	Pending [OI-02]
CON-05	Non-Refundable 25% Deposit Disputes	Nagwa vs. Citizens	Instant spatial pre-screening runs *before* payment + signed digital disclaimer.	Resolved [OI-06]
CON-06	Unpriced Extensions & Street Rates	Nagwa vs. Pricing	Street rate overrides zone; unpriced extensions halt as `AWAITING_PRICING`.	Resolved [OI-05]
CON-07	Live Civil Registry API Lead Time	Walid vs. Sponsor	Phase 1 uses clerk ID check + regex; live ministerial API deferred to Phase 2.	Scoped [DEP-02]
CON-08	Utility Meter Crew Dispatch Request	Khaled vs. Scope	Crew scheduling excluded (`OOS-01`); delivered read-only REST Verification API.	Enforced [A-03]
CON-09	38k Active Backlog vs. 40k Archive	Sayed vs. Walid	Phase 1 migrates 38k active backlog; closed archive scanned on active appeal.	Scoped [OI-10]
CON-10	Red Belt vs. WGS 84 Datum Shift	Ahmed vs. Accuracy	Standardized on EPSG:32636; ETL applies certified Survey Authority shift grid.	Pending [OI-04]

Table 20: Stakeholder Conflict Resolution Log

Assumptions, Constraints & Dependencies: **A-01:** On-premises hosting in Governorate DC (Oracle 11g / ArcGIS 11.1); **A-02:** Unified DB for portal and clerk intake; **A-03:** Utility integration limited to read-only REST API; **A-04:** Offline tablet autonomy via SQLite; **A-05:** Algorithmic UPIDs for un-cadastered rural properties; **A-06:** All tariffs, timers, and rules are parameter-driven without code changes. **Constraints:** Workforce fixed at 25 engineers; unstable village internet; quarterly satellite delivery. **Dependencies: DEP-01:** Quarterly imagery supply; **DEP-02:** MOI Civil Registry API approval (Phase 2); **DEP-03:** Survey Authority datum parameters; **DEP-04:** Extension zone pricing decrees.

ID	Risk Description	P	I	S	Proactive Architectural & Governance Mitigation
RSK-01	86.7% Committee Throughput Deficit	5	5	**25**	Deploy spatial pre-screening for desk triage [OI-02]; escalate session scaling ($\sim$20/wk) [OI-01].
RSK-02	Rural Internet Outages Halting Fieldwork	4	4	**16**	Offline-first tablet app with cached vector tiles, SQLite, and background delta-sync (`FR-FLD-001`).
RSK-03	Litigation from Premature Demolition	4	4	**16**	Enforce 30-day appeal timer and immutable `STAY_OF_EXECUTION` database lock (`FR-COM-005`).
RSK-04	Disputes Over Non-Refundable 25% Fee	4	3	**12**	Run instant spatial prohibition check prior to payment voucher + signed disclaimer (`BR-VAL-004`).
RSK-05	Datum Shift Causing False Rejections	3	4	**12**	Standardize on WGS 84 / UTM 36N using certified Survey Authority shift parameters [OI-04].
RSK-06	Corrupted/Duplicate Legacy Excel Data	4	3	**12**	Automated ETL deduplication on National ID/coordinates with clerk quarantine queues (`FR-MIG-002/004`).
RSK-07	Utility Scope Creep into Crew Dispatch	3	3	**9**	Enforce `OOS-01` boundary; provide read-only REST Verification API (`FR-UTL-003`).

Table 21: Enterprise Risk Register (P: Probability 1–5, I: Impact 1–5, $S = P \times I$)

16. Appendices

16.1. Appendix A: Comprehensive Glossary and Acronyms

Table 22 defines the primary technical, geospatial, legal, and operational terms governing the BVDRS platform.

Term	Expansion	Operational Definition in BVDRS
BVDRS / UPID	System / Unique Property ID	Al-Wadi reconciliation platform / Geocode (`GOV-DIST-ZONE-GRID-SEQ`) indexing un-cadastered buildings.
WGS 84 / Red Belt	Geodetic Coordinate Systems	Target standard EPSG:32636 (WGS 84 / UTM 36N) vs. legacy Egypt 1907 Red Belt (EPSG:22992).
Fast-Track Triage	Expedited Committee Routing	Spatial pre-screening routing zero-conflict urban files to committee desk review (toggle OFF [OI-02]).
Stay of Execution	`STAY_OF_EXECUTION` Lock	Automated DB lock blocking demolition orders during the 30-day appeal window and appellate review.
POA / OLS	Power of Attorney / Aviation	Notarial proxy for legal agent filings / Airport Obstacle Limitation Surfaces restricting building height.

Table 22: Appendix A: Core Project Glossary and Technical Acronyms

16.2. Appendix B: Comprehensive Open Issues Log

Table 23 catalogs the ten unresolved governance, policy, and technical issues, establishing governing authorities and interim architectural baselines.

ID	Open Governance Issue	Owner	Imp.	Interim Architectural Baseline / Fallback Position
OI-01	6-Month Target vs. 86.7% Deficit	Governor; Hesham	Crit	Triage deployed; recommended $\sim$20 extra committee sittings/wk.
OI-02	100% Inspection vs. Desk Triage	Mona; Legal	High	100% visit is default; Fast-Track desk triage built as configurable toggle (OFF).
OI-03	Weekly Satellite Imagery Budget	Hesham; Ahmed	High	Baselined on quarterly 50 cm imagery contract; Phase 2 drone support.
OI-04	Datum Shift Parameters	Survey Auth; Ahmed	High	Standardized on EPSG:32636; target error ≤ 0.3 m [TBC] pending certification.
OI-05	Unpriced Urban Extension Tariffs	Pricing Cmte; Nagwa	High	Status held as `AWAITING_PRICING`; system prohibits fee guessing.
OI-06	25% Deposit Refund & Arrears Rules	Nagwa; Legal	Med	Pre-screening precedes payment + disclaimer; $T + 30/60$ arrears configurable.
OI-07	30-Day Appeal Timer Start Trigger	Mr. Adel (Legal)	High	Initiates upon confirmed SMS/electronic delivery timestamp [TBC].
OI-08	Certificate Trigger on Instalments	Nagwa; Legal	High	Issuance trigger parameterized (after 25% deposit vs. full settlement [TBC]).
OI-09	Civil Registry National Portal API	Walid; MOI	Med	Phase 1 uses standalone portal + clerk ID check; live APIs in Phase 2.
OI-10	38k Backlog vs. 40k Paper Archive	Sayed; Walid	Med	Phase 1 prioritizes 38k active pending files; closed archive scanned on appeal.

Table 23: Appendix B: Open Issues Log and Interim Baselines

16.3. Appendix C: Traceable User Stories (Gherkin Format)

Table 24 specifies five primary personas across the BVDRS lifecycle with testable `Given / When / Then` acceptance criteria.

ID & Persona	User Story	Testable Acceptance Criteria (Given / When / Then)
US-01 Citizen / Legal Agent	Submit application online, pin building, and pay 25% deposit.	**Pass:** Given valid docs & pin outside prohibition zones; When 25% paid; Then UPID & receipt issue. **Block:** Given pin in Nile buffer; When submitted; Then halts payment & flags ineligible.
US-02 Tech Center Clerk	Log walk-in paper files and POAs into the central database.	**Assisted Intake:** Given walk-in citizen/lawyer with POA; When clerk validates ID/POA and pins map; Then dossier saves to central DB and prints barcoded receipt.
US-03 Field Engineer	Inspect rural sites offline using cached basemaps and GPS camera.	**Offline Sync:** Given tablet with zero signal; When engineer adjusts GPS vertices, logs floors, and takes in-app photo; Then report stores in SQLite and auto-syncs within 60 s of Wi-Fi/4G.
US-04 Committee Member	Adjudicate digital dockets with spatial summaries and locks.	**Approve/Reject:** Given docket reviewed; When panel votes Approve, routes to billing; When panel votes Reject, starts 30-day timer and applies `STAY_OF_EXECUTION` lock.
US-05 Utility Engineer	Verify certificates online via REST API or QR scan before meter install.	**API Check:** Given QR/UPID queried; When certificate is active, returns *ACTIVE* + floors in ≤ 3 s; When account is > 60 days in default, returns *SUSPENDED* and blocks meter.

Table 24: Appendix C: Traceable User Stories and Gherkin Scenarios